

BoxCraft: Tangible Mixed Reality for CAD Modelling

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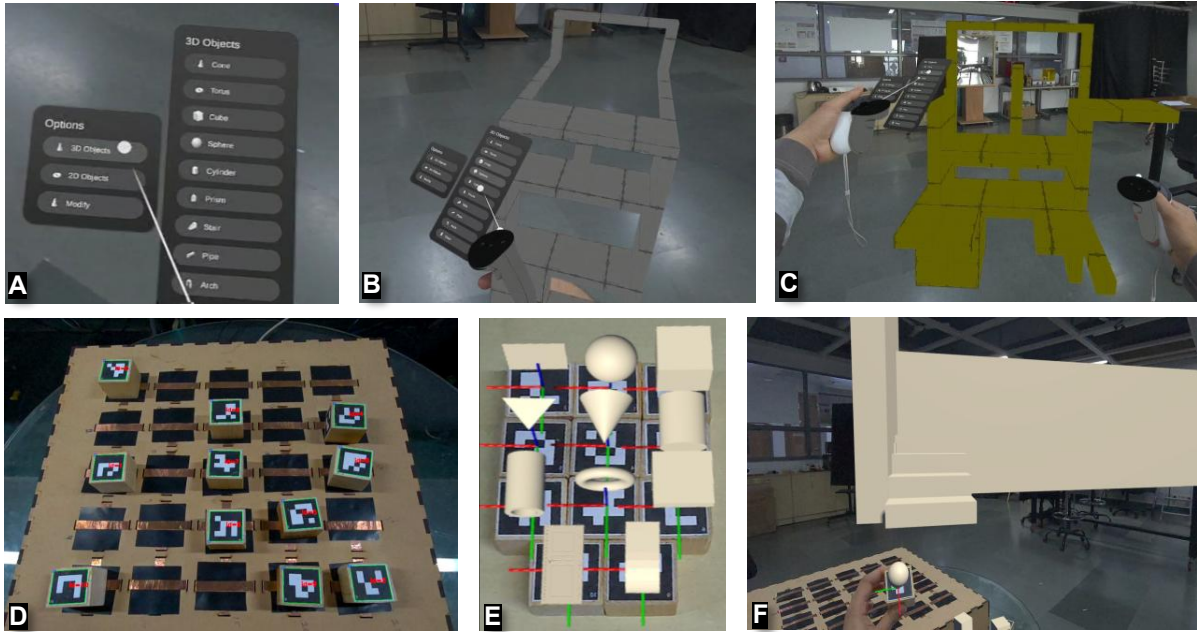


Figure 1: BoxCraft tangible mixed reality (MR) system with two modes: VR controller-based interaction (A-C) and tangible token-based interaction (D-F). Controller-based shape creation is shown in (A-C), while (D-F) show token tracking and 3D model interaction in MR.

ABSTRACT

BoxCraft is a tangible mixed reality system to support beginners in 3D CAD modelling through phygital interaction. The system has two interfaces: VR controller and tangible token-based interface to build and manipulate 3D models. We conducted a participatory design workshop ($n=6$) for gathering early design insights and then ran a comparative user study ($n=7$) to measure task performance using SUS and Raw NASA-TLX. The controller-based interface allowed for faster completion of tasks with lower workload, although the tangible token-based interface reduced CAD complexity and offered higher conceptual clarity at an early stage, making it potentially useful for early-stage CAD modelling.

Index terms: Mixed Reality, Tangible User Interface, Computer-Aided Design, Tangible Extended Reality.

1 INTRODUCTION

CAD design using traditional CAD systems is often a challenging task for new learners [1]. To address this, virtual reality-based CAD

systems were introduced for immersive modelling, but these systems primarily depend on mid-air or controller-based input [2], and due to limitations in accuracy, they have seen limited adoption in academia and industry [3]. One promising direction explored in prior work is the integration of extended reality (XR) with tangible user interfaces (TUI), which can support alternative interaction paradigms for CAD tasks, such as MagicBook [4], Tesla Block [5], and Tangible Globes [6]. Prior studies suggest that TUI can reduce initial learning barriers for first-time XR users by leveraging familiar physical interactions [7].

2 METHODOLOGY

2.1 System design

To address these challenges, we propose BoxCraft, a system that combines TUI with MR for 3D CAD modelling. The system supports two distinct interaction modes: a controller-based interface using a Meta Quest 3 controller, and a tangible token-based interface in which users interact with a physical cube token to create and modify 3D primitive models. ArUco markers placed on the cube enable tracking, and the corresponding 3D model is overlaid on the token, visualised in MR space.

The tangible interface consists of a 5×5 velostat sheet grid connected to an Arduino board. Each grid cell functions as an input region for the MR environment. Placing the cube on a grid defines the placement of the 3D model in the MR space, on applying pressure controls, continuous scaling of the primitive geometric shapes are observed. Interaction data is streamed to an application built using Unity, where the ProBuilder API is used to generate and modify 3D models in MR.

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2.2 Participatory Design Workshop

We conducted a participatory design workshop involving 6 participants (3F, 3M) with diverse backgrounds. Participants had limited prior experience with MR and beginner-level familiarity with CAD tools. The workshop elicited early design feedback for BoxCraft. Participants engaged with the controller-based interface, followed by the tangible token-based interface, after which feedback was collected through group discussion.

2.3 User Study

We conducted a user study with 7 participants (3F, 4M). To minimize familiarity bias and get an objective usability measure, different participants were used for the workshop and the user study. Participants were given a one-minute demonstration of both interfaces, followed by time for free exploration. Participants then completed the tasks (T1-T3; Table 1) with both interfaces so that their performance could be directly compared.

Table 1. Representation of tasks performed during the user study.

Task no.	Task performed	What it tests
T1	Create any shape	Basic usability
T2	Scale the 3D model	Interaction accuracy
T3	Modify the shape	Spatial manipulation

During task execution, we recorded the frequency of errors and instances where participants required assistance operating the system as a metric of system usability and learnability during the user study. After completing the task, participants filled out the System Usability Scale (SUS) and the Raw NASA-TLX to assess usability and perceived workload.

3 RESULTS AND DISCUSSION

We used a participatory design process, including a hands-on workshop and a comparative user study. Participants found the geometric shapes familiar and suggested clearer visual feedback, such as outlines, labels, and color distinction. In terms of interaction techniques, participants preferred straightforward and explicit controls (e.g., sliders, knobs, buttons, single-tap gestures) to reduce cognitive load.

Assessment outcomes revealed that participants completed CAD modelling tasks sooner with the controller compared to the tangible token-based interface (see Fig. 2). The tangible token-based interaction resulted in more errors than the controller-based interaction and required more assistant interventions (see Fig. 3).

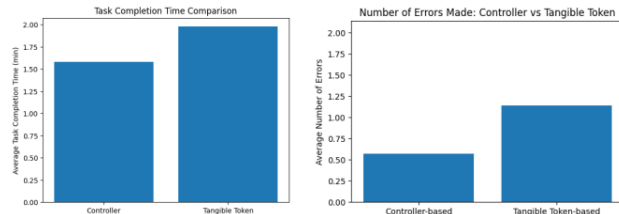


Figure 2: Task completion time

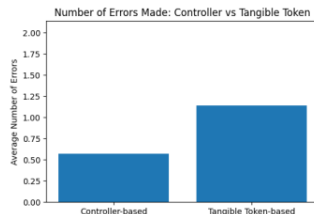


Figure 3: Number of errors

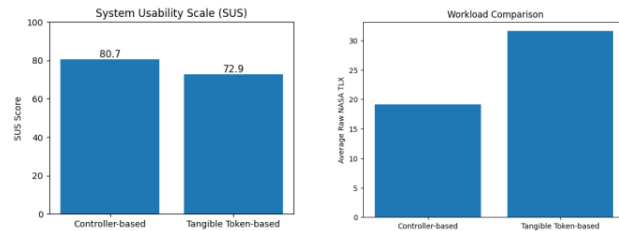


Figure 4: SUS results

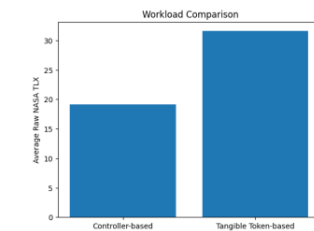


Figure 5: Raw NASA-TLX

SUS scores demonstrated that the usability of both interaction modalities was good to excellent, although differences between the controller-based and tangible token-based systems were notable (see Fig. 4). Finally, participants rated their overall workload for the tangible token-based interaction higher than for the controller-based interaction (see Fig. 5).

The controller-based interaction showed the ability for swift task execution and low workload, while the tangible token-based system revealed good usability and was perceived to support users' understanding of basic shape composition and spatial relationships. The high workload of the tangible interface may be due to the lack of familiarity with the physical token. Participants reported that the tangible token supported conceptual reasoning about complex shapes and basic CAD concepts, but would benefit from more clear physical affordances, and stronger visual cues to help reduce errors and cognitive overload. Together, the above results indicate that tangible systems like BoxCraft are potentially useful for early-stage CAD modelling tasks, where object translation, rotation, and conceptual understanding are valued more than speed.

4 CONCLUSION AND FUTURE WORK

This paper presents BoxCraft, a system utilizing TUI and MR for CAD modelling. A participatory design workshop and a comparative user study were conducted to evaluate the system. The results show that a controller-based interface allowed for faster task performance with lower workload, while the tangible token-based interface provided decent usability and supported conceptual understanding for novice users. Overall, BoxCraft shows potential to enable intuitive and engaging CAD modelling experiences in mixed reality.

In future work, we will refine the tangible token design with explicit physical affordances and enhanced visual feedback. Future work also includes larger user studies to compare learning outcomes and alternative tangible form factors from the cube to support more complex CAD modelling and CAD education.

5 ACKNOWLEDGMENT

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REFERENCES

- [1] S. Lee-Remond, S. Sagot, and E. Ostrosi, "A New Design Framework for Comprehensible Graphical User Interfaces for Parametric Computer-Aided Design Tools," *Comput. Aided. Des. Appl.*, pp. 150–179, Aug. 2024, doi: 10.14733/cadaps.2025.150-179.
- [2] S. M. Feeman, L. B. Wright, and J. L. Salmon, "Exploration and evaluation of CAD modeling in virtual reality," *CAD Solutions LLC*, vol. 15, no. 6, pp. 892–904, Nov. 2018, doi: 10.1080/16864360.2018.1462570.
- [3] T. T. H. Tran, G. Foucault, and R. Pinqu  , "Benchmarking of 3D Modelling in Virtual Reality," *Comput. Aided. Des. Appl.*, vol. 19, no. 6, pp. 1184–1190, 2022, doi: 10.14733/cadaps.2022.1184-1190.
- [4] M. Billingham, H. Kato, and I. Poupyrev, "The MagicBook: a transitional AR interface," 2001.
- [5] K. Ikegawa and B. Shizuki, "Tesla blocks: Magnetism-based tangible 3d modeling system using block-shaped objects," in *ACM International Conference Proceeding Series, Association for Computing Machinery*, Dec. 2018, pp. 411–415. doi: 10.1145/3292147.3292221.
- [6] K. A. Satriadi et al., "Tangible Globes for Data Visualisation in Augmented Reality," *Conference on Human Factors in Computing Systems - Proceedings*, Apr. 2022, doi: 10.1145/3491102.3517715/SUPPL_FILE/3491102.3517715-VIDEO-PREVIEW.MP4.
- [7] K. Wang, P. Liu, and K. Perlin, "Exploring Tangible and Graphical Interfaces for Large Virtual Object Manipulation," *Association for Computing Machinery (ACM)*, May 2025, pp. 1–2. doi: 10.1145/3722564.3728389.